

Step 5: Live Orbit Tracker

Real-time satellite positions and physics-based overpass predictions — no Earth Engine needed.

This step pulls each satellite's current real orbital elements (TLEs) from CelesTrak, a free public catalog, and propagates them forward using the same orbit-mechanics technique used by sites like N2YO. It needs no Earth Engine project and no login of any kind.

What it predicts, and what it doesn't

This predicts **when a satellite's swath geometrically crosses your AOI** — a necessary condition for an image, but not a guarantee one was taken:

- **Sentinel-2 and Landsat** acquire systematically on every qualifying pass, so these predictions are directly reliable.
- **Sentinel-1** is tasked — whether ESA actually collects an image on a geometrically valid pass is a separate planning decision this cannot see. Useful as a cross-check against the Forecast step, especially when that step has too little history for a confident pattern.

Reading the map

"Current position" reflects the moment you clicked Refresh — it does not update live on its own; click Refresh again to get a new snapshot. Each satellite's ground track line spans roughly 20 minutes before to 100 minutes after that moment, so you can see which direction it's heading.

■ **Tip** — Swath widths used here are approximate nominal figures (Sentinel-1 IW $\approx$ 250 km, Sentinel-2 $\approx$ 290 km, Landsat $\approx$ 185 km) — real swaths vary slightly by acquisition mode.